

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

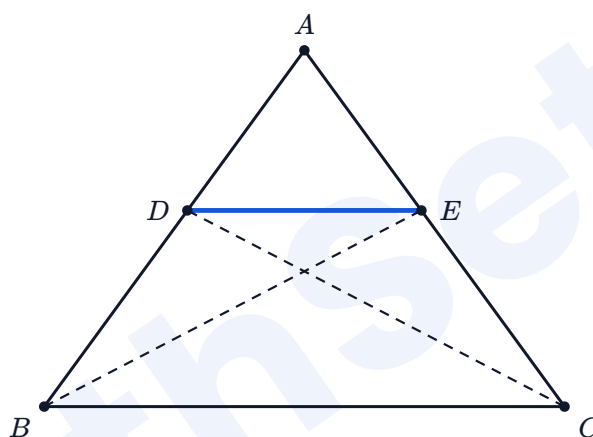
# Chapter 6: Triangles

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

## 1 Basic Proportionality Theorem

1. If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio. **CBSE 2024, 5 marks**



### SOLUTION

Given: in  $\triangle ABC$ ,  $DE \parallel BC$ , with  $D$  on  $AB$  and  $E$  on  $AC$ . To prove:  $\frac{AD}{DB} = \frac{AE}{EC}$ .

Construction: join  $BE$  and  $CD$ . Draw  $EN \perp AB$  and  $DM \perp AC$ .

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN} = \frac{AD}{DB}$$

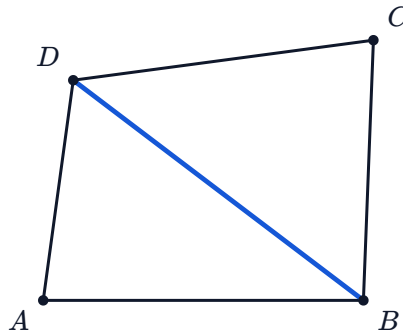
$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle CED)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \frac{AE}{EC}$$

$\triangle BDE$  and  $\triangle CED$  are on the same base  $DE$  and between the same parallels  $DE$  and  $BC$ , so  $\text{ar}(\triangle BDE) = \text{ar}(\triangle CED)$ .

Hence  $\frac{AD}{DB} = \frac{AE}{EC}$ .

## 2 Criteria for similarity of triangles

2. In the given figure,  $ABCD$  is a quadrilateral. Diagonal  $BD$  bisects  $\angle B$  and  $\angle D$  both. Prove that (i)  $\triangle ABD \sim \triangle CBD$  (ii)  $AB = BC$ . **CBSE 2024, 2 marks**



### SOLUTION

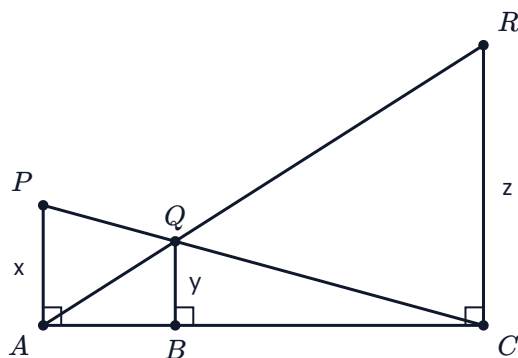
(i) In  $\triangle ABD$  and  $\triangle CBD$ :  $\angle ABD = \angle CBD$  ( $BD$  bisects  $\angle B$ ) and  $\angle ADB = \angle CDB$  ( $BD$  bisects  $\angle D$ ).

So  $\triangle ABD \sim \triangle CBD$  by the AA criterion.

(ii) Corresponding sides are proportional:  $\frac{AB}{CB} = \frac{BD}{BD} = 1$

**Hence**  $AB = BC$ .

3. In the given figure  $PA$ ,  $QB$  and  $RC$  are each perpendicular to  $AC$ . If  $AP = x$ ,  $BQ = y$  and  $CR = z$ , then prove that  $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$ . **CBSE 2024, 5 marks**



**SOLUTION**

In  $\triangle CQB$  and  $\triangle CPA$ :  $\angle C$  is common and  $\angle CBQ = \angle CAP = 90^\circ$ , so  $\triangle CQB \sim \triangle CPA$  (AA).

Hence  $\frac{BQ}{AP} = \frac{BC}{AC}$ , that is  $\frac{y}{x} = \frac{BC}{AC}$ .

In  $\triangle AQB$  and  $\triangle ARC$ :  $\angle A$  is common and  $\angle ABQ = \angle ACR = 90^\circ$ , so  $\triangle AQB \sim \triangle ARC$  (AA).

Hence  $\frac{BQ}{CR} = \frac{AB}{AC}$ , that is  $\frac{y}{z} = \frac{AB}{AC}$ .

Adding:  $\frac{y}{x} + \frac{y}{z} = \frac{BC + AB}{AC} = \frac{AC}{AC} = 1$

Dividing by  $y$ :  $\frac{1}{x} + \frac{1}{z} = \frac{1}{y}$ .